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Chinese Solar Farms Are Crowding Out Much-Needed Crops

As China expands renewable energy at a furious pace, solar farms are encroaching on cropland

By *Chun Han Wong* [Follow](#)

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China installed more solar-power capacity last year than the U.S. has built in its history. Now Beijing is worried that the push may have gone too far in some places as solar farms encroach on cropland, undermining leader Xi Jinping's goal of ensuring China can feed itself.

Backed by soaring demand for renewable energy, solar-power projects have become lucrative enough—especially when state subsidies are included—that some companies, local officials and farmers are trying to cash in by repurposing areas once dedicated to crops, defying Beijing's diktats against developing arable land.

The issue garnered national attention after state broadcaster China Central Television aired a report on it earlier this year.

In the rural township of Muzi in Hubei province, a major grain-producing region, CCTV found that several hundred acres once earmarked as “high-quality farmland” had been covered with solar photovoltaic panels, even though local authorities had announced plans in 2019 to build irrigation channels, drainage and roads there to boost crop yields and improve connectivity.

While the solar sector deserves support, “no matter how good the industry, it shouldn't violate state laws and go against the central government's policies,” CCTV warned. “The protection of farmland is a major matter related to national security strategy.”

Campaigns collide

The exposé, and other similar cases, have highlighted how Xi's sweeping priorities can sometimes clash on the ground.

Food security is paramount for Chinese leaders, given the country's limited water supply and arable land. Past food shortages including the Great Famine of 1959-61 threatened stability, and climate change is raising fears of new threats to agriculture, with drought affecting parts of the country this year. Despite some increases in food production in recent years, it hasn't been enough to keep up with increasing demand.



Climate change is raising concerns about threats to agriculture and the loss of arable land in China. PHOTO: CFOTO/DDP/ZUMA PRESS

Xi has said that officials must “resolutely defend China’s arable land red line” of roughly 300 million acres nationwide to ensure the country doesn’t become more dependent on imports, including soybeans from the U.S. and elsewhere.

According to state media, the rate at which arable land has disappeared due to urbanization, illegal encroachment and other factors has accelerated in recent decades. The government’s latest land survey showed China lost more than 18 million acres of arable land in the 10 years up to the end of 2019, bringing the total down to roughly 316 million acres.

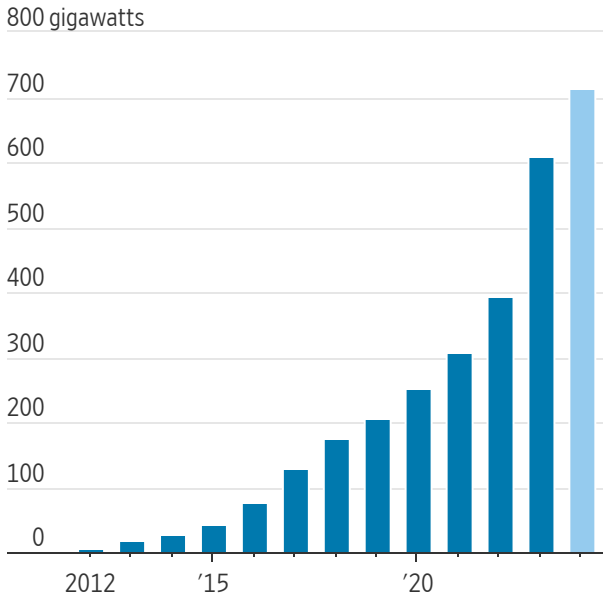
“As early as 2013, I’ve said that we must protect farmland like how we protect giant pandas,” Xi said in 2020.

But Xi has also called on local officials to promote renewable energy as part of China’s goals of cutting carbon emissions and reducing its dependence on imported oil and coal. Government support includes subsidies, cheap loans and tax incentives for solar and wind businesses that range from small startups to major listed companies and state enterprises.

While there is no official data on how much land is being swallowed up by solar farms, government data tracked the addition of nearly 217 million kilowatts of new solar capacity

Shining Brighter

China's grid-connected solar power capacity



Note: Data for 2024 is as of June.

Source: China's National Energy Administration

last year, up 55% from 2022.

China accounted for more than half the global increase in solar capacity in 2023, according to energy research firm Wood Mackenzie. Analysts estimate China will need to expand its solar-generation capacity some 14 times from 2020 levels to realize Xi's target of carbon neutrality by 2060.

Other countries, including the U.S., have also seen solar operations taking over farmland—a problem that dates back years.

A recent study by Chinese researchers estimated that in 2018, food-production losses resulting from solar-power facility development on cropland globally reached an amount sufficient for feeding 4.3 million people for a year, with a third of these losses

attributable to China's solar-power surge.

“As solar energy expands, the loss of cropland will become a pressing issue that requires urgent attention,” another group of researchers warned in a November commentary published by the journal Nature Geoscience. “Countries grappling with renewable energy development and facing spatial development imbalances, as in China, must be mindful of the potential conflicts between energy and land.”



China accounted for more than half the global increase in solar capacity in 2023. PHOTO: CFOTO/DDP/ZUMA PRESS

Competing for land

While solar-power developers run into fewer conflicts with agriculture in western China, which has more open spaces that aren't ideal for farming due to low rainfall, their expansion across eastern China puts them more directly in competition with farming.

“There is a lot of money to be made from renewables, while agriculture is a low-value and low-profit business,” said Cosimo Ries, a renewable-energy analyst at the consulting firm Trivium China.

Officials have tried to balance the competing demands by promoting *nongguang hubu*, or “agriculture and solar complementing each other.” The idea involves encouraging “agrivoltaic” facilities in which solar panels are installed in ways that allow the land to still be used for farming or grazing.

According to one Chinese academic paper, grid-connected dual-purpose agrivoltaic projects accounted for about 7% of China's photovoltaic capacity as of 2019.

The idea has also caught on in parts of Europe and the U.S., including California. But experts say the presence of solar panels limits what can be grown, with crops such as lettuce and broccoli tending to do better than grains that require lots of sunlight.

In practice, experts say, some Chinese solar farm developers fail to dedicate much attention to crops. Local governments, beset with heavy debts amid a nationwide real estate bust, often look the other way so long as solar developers bring investments, tax revenues and jobs.

They are “eagerly selling and leasing land in return for immediate and temporary benefits,” Zhu Qizhen, a professor at China Agricultural University, told a state media outlet.



Farmers in Linyi, China, tend to crops planted under solar panels at an agrivoltaic farm. PHOTO: WANG YANBING/VCG/GETTY IMAGES

Government clampdown

While Beijing still offers subsidies for solar projects, it has been punishing companies and officials it believes are exploiting government support at agriculture's expense.

Beijing issued a directive last year stipulating that solar photovoltaic projects shouldn't be built on farmland, grassland or protected forest land, prompting at least 10 regional governments to publish new or updated rules on solar-power land use. The directive followed moves in 2021 in which several provinces halted new photovoltaic projects pending reviews to assess whether they improperly encroached onto arable land.

In October, China's Ministry of Natural Resources reprimanded authorities in the northern county of Huanglong for failing to prevent a renewable-energy firm from illegally installing solar panels on agricultural land. Authorities in the southern province of Guizhou in January ordered a local company to pay fines worth nearly \$100,000 for illegally building a solar-power facility on communal farmland.

In the case of Hubei, CCTV aired aerial footage showing a sprawling array of solar panels occupying acres of land in Muzi township that residents say was previously used for growing rice. Residents said they were persuaded to lease the land to a Hong Kong-listed solar-farm company for an agrivoltaic project, but the facility's operator only made superficial efforts to grow rice—which isn't suited for growing under shade—and produced far lower yields than before.

Local farmers told CCTV that the area's agricultural yield had fallen by some 80%.

When approached by CCTV, a worker at the agrivoltaic facility said there was no protected cropland in the vicinity, a claim contradicted by local residents and signage stating that the area was designated “high quality” farmland reserved for agriculture. The solar company didn’t immediately respond to a request for comment.

CCTV uncovered similar problems elsewhere in Hubei, including the township of Huaxi, where large fields of solar panels have been installed on farmland. A local land-resources official told CCTV that authorities had redesignated it as ordinary land, and compensated by earmarking other plots for food production.

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